L05 Two ways to add a resistor

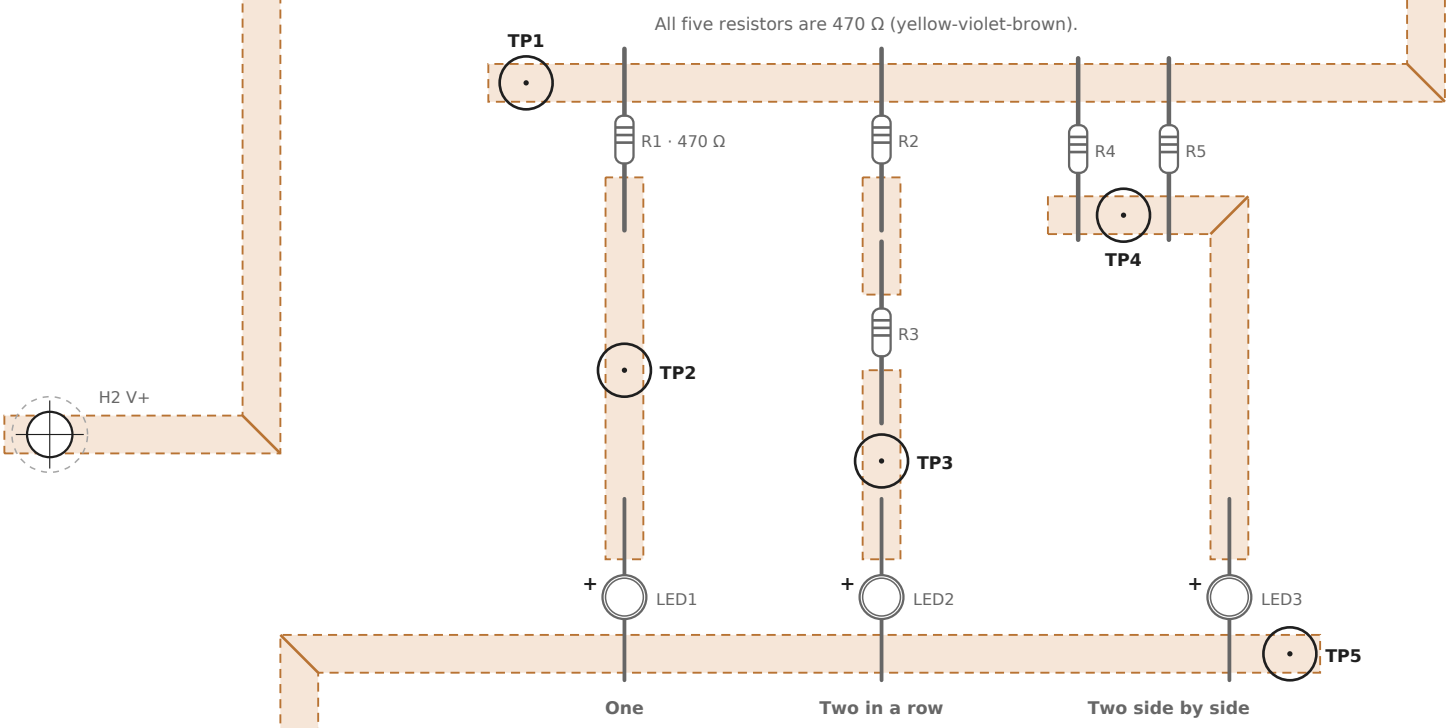
L05

One resistor, two in a row, two side by side. Which LED shines brightest?

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down



H2 V+

H3 GND

You need

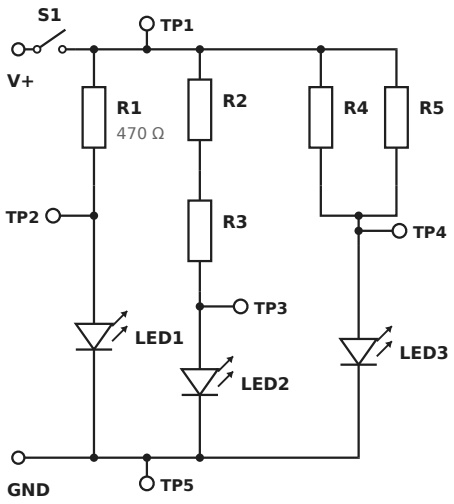
- 3 red LEDs, 5 mm
- 5 resistors, 470 Ω (yellow-violet-brown) **[P]**
- 5 mm copper tape
- Power page P1 in the binder
- A multimeter for Measure

Build it

1. Lay the seven lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Place R1 to R5 across their gaps, either way round. R4 and R5 share one gap, side by side.
5. Place LED1, LED2 and LED3 as printed: long leg on the +.
6. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
7. Bind this page with P1, then fold the corner down to switch on.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Schematic



R1 to R5 are all 470 Ω.

Predict Rank LED1, LED2 and LED3 from brightest to dimmest.

Brightest _____ _____ _____ **Dimmest**

Which group of resistors do you think lets the most current through?

Measure Switch BT1 off. Measure from TP1 to each point.

From TP1 to	Ohms	
TP2 one		Ω
TP3 two in a row		Ω
TP4 two side by side		Ω

Then switch on.

Did your ranking match?

Explain Two in a row add up.

What do two side by side do, and why does that LED shine brightest?

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.